# Case 32: Raster-orientation covariance

## 1 Verification purpose

This case verifies that ELMFIRE's raster indexing and level-set transport are covariant under a 90° rotation. It is the portable successor to Adam Laird's orientation of the prescribed input fields scenario.

## 2 Mathematical formulation and numerical implementation

For homogeneous isotropic forcing, the operator $\mathcal{L}$ in $\partial_t \phi = \mathcal{L}(\phi)$ commutes with any rigid rotation $Q$: $\mathcal{L}(\phi \circ Q^{-1}) = (\mathcal{L}\phi) \circ Q^{-1}$. A square north-up raster rotated by exactly 90 degrees requires no interpolation, so cell-by-cell comparison is available.

## 3 Derivation of expected behavior

The rotated simulation condition is generated with a rotation through a multiple of 90 degrees from the asymmetric L-shaped initial negative-$\phi$ region. Fuel, moisture, slope, and wind are spatially uniform. Therefore the rotated arrival-time validity mask and values must equal a 90-degree rotation of the original result, apart from bounded grid-level roundoff.

The reference array is generated directly from the original ELMFIRE arrival-time result by a 90-degree counterclockwise rotation, $T_{\text{ref}} = \text{rot90}(T_{\text{original}})$; no arrival values are invented. With $V = \{i : T_{\text{ref},i} \text{ and } T_{\text{rot},i} \text{ finite}\}$ and $U = \{i : \text{either value is finite}\}$, finite-mask agreement is

$$A_m = \frac{|V|}{|U|},$$

whose exact rotational reference is 1 and whose substituted test is $A_m \geq 0.99$. On the mutual finite mask, the value metric is

$$E_T = \frac{|V|^{-1} \sum_{i \in V} |T_{\text{ref},i} - T_{\text{rot},i}|}{\max[P_{90}(T_{\text{ref},i} : i \in V), 1 \text{ s}]},$$

with exact reference zero and acceptance substitution $E_T \leq 0.02$. The asymmetric generated ignition mask makes this comparison sensitive to an erroneous reflection or transposition.

## 4 Verification metrics and acceptance criteria

Finite-mask intersection over union must be at least 0.99. On mutually finite cells, normalized mean absolute arrival-time error must be at most 0.02. Mask agreement detects row/column reversal and georeferencing errors; value error detects orientation-dependent numerical evolution.

## 5 Simulation configuration and predicted outcome

Both simulation conditions use a 2000 m square domain at 5 m, Fuel Model 10, flat terrain, zero wind, and identical timing. The only changed scientific input is the exact raster rotation of level-set field. The second result is predicted to be the corresponding rotation of the first.

| Parameter | Configured value | Units | Role |
|---|---|---|---|
| Domain/grid | 2000 m square; 5 | m/cell | Exact square raster rotation |
| Fuel/forcing | FM10; flat; calm | — | Isotropic homogeneous operator |
| Time control | step 1, maximum 5; stop 7200 | s | Common exact comparison time |
| Initial field | asymmetric L shape | — | Makes orientation errors observable |
| Simulation condition transform | 90-degree rotation | degrees | Exact no-interpolation covariance test |

Figure 1 shows the prescribed spatial condition used in this experiment. The displayed field represents the initial level-set geometry or a controlling fuel, terrain, moisture, wind, or canopy quantity, as identified by the figure labels. The plotted values come from the prepared simulation inputs, not from the expected response.

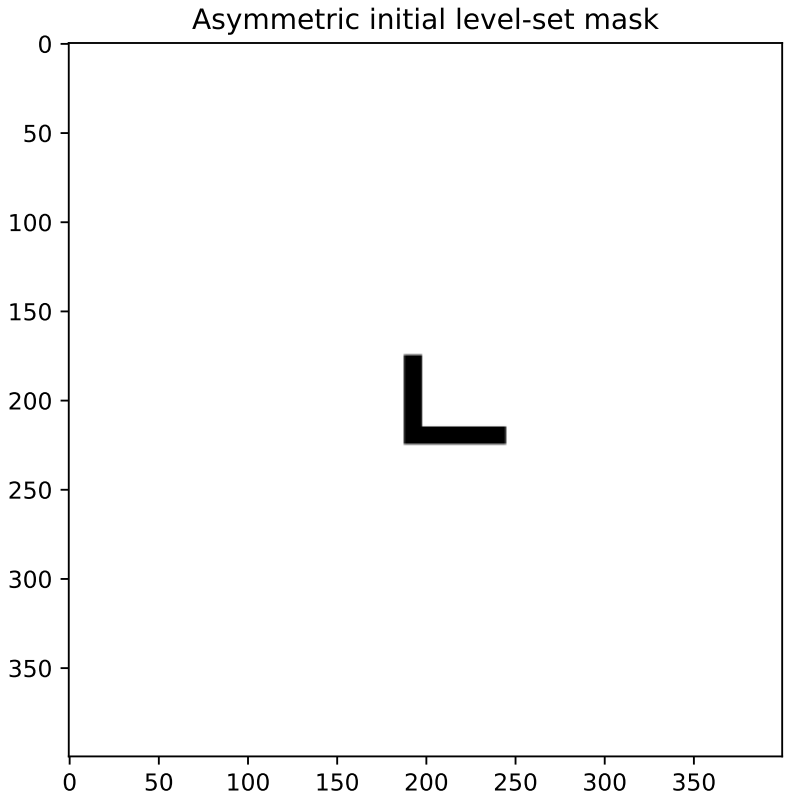



Figure 1: Prescribed asymmetric initial level-set geometry. Coordinates are grid-cell indices on the 5 m grid; the dark region identifies the initial ignition geometry.

## 6 Actual simulation results

The generated status is **PASS**; 2 of 2 metrics pass across 2 simulation conditions. Missing output produces NOT EVALUABLE.

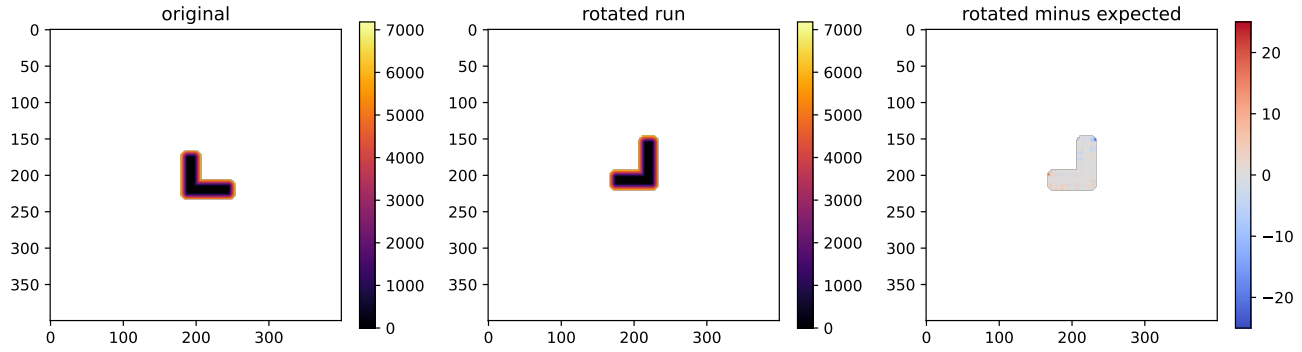



Figure 2: Original, independently rotated, and difference arrival-time fields.

## 7 Calculated metrics and verification decision

| Metric | Expected | Calculated | Status |
|---|---|---|---|
| rotated finite-mask agreement | >= 0.99 (1% disagreement) | 1 fraction | PASS |
| rotated arrival-time error | <= 0.02 (2%) | 0.000214081 normalized L1 | PASS |

The authoritative decision is **PASS** from calculated verification and validation metrics.

## 8 Technical interpretation

Mask disagreement indicates inconsistent raster orientation, bounds, or row/column indexing. Value disagreement with matching masks indicates a directional bias in transport or output serialization.

## 9 Scope and limitations

The 90-degree test avoids resampling and therefore does not assess arbitrary-angle reprojection. It verifies executable behavior, not the internal bilinear-interpolation routine in isolation.